NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 2.

SATURDAY 23RD JUNE, 2018 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are two sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*).*

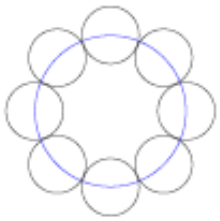
(c) *Marks will be awarded for only answers for which a clear and logical layout of the working is provided. A correct solution, poorly presented, will not earn full marks.*

(d) ***Indicate your names, class, school and district on all your answer sheets**.*

(e) *Answer scripts of only qualified students should be returned to the contest coordinator, latest Monday 25th June, 2018.*

(f) *ALL participants who qualified for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 28*th July,* 2018 *at 8:00am at Makerere University, Kampala.*

SECTION A : 5 marks each



Qn. 1. Eight circles of radius 1cm have centers on a larger common circle and adjacent circles are tangent. Find the area of the common circle.

Qn. 2. Determine all the real solutions of the following equation:

$$2^{(2^x)} - 3 \cdot 2^{\left(2^{x-1}+1\right)} + 8 = 0.$$

Qn. 3. Compute the integer k, $k > 2$, for which

$$\log_{10}[(k-2)!] + \log_{10}[(k-1)!] + 2 = 2\log_{10}(k!).$$

Qn. 4. Let N be a 3-digit number with three distinct non-zero digits. We say that N is *mediocre* if it has the property that when all six 3-digit permutations of N are written down, the average is N. For example, $N = 481$ is mediocre, since it is the average of $418, 481, 148, 184, 814, 841$. Determine the largest mediocre number.

Qn. 5. Prove that for all real numbers x, y,

$$\left(x^2+1\right)\left(y^2+1\right) + 4\,(x-1)\,(y-1) \geq 0.$$

Determine when equality holds.

Qn. 6. If

$$x = \left(1 + \frac{1}{n}\right)^n \quad \text{and} \quad y = \left(1 + \frac{1}{n}\right)^{n+1},$$

show that $y^x = x^y$.

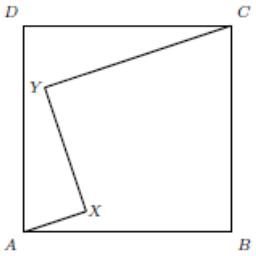
Qn. 7. Find all number triples (x, y, z) such that when any of these numbers is added to the product of the other two, the result is 2.

Qn. 8. Suppose that n people each know exactly one piece of information, and all n pieces are different. Every time person A phones person B, A tells B everything that A knows, while B tells A nothing. What is the minimum number of phone calls between pairs of people needed for everyone to know everything?

Qn. 9. Simplify

$$\left(\frac{1 \cdot 2 \cdot 4 + 2 \cdot 4 \cdot 8 + \ldots + n \cdot 2n \cdot 4n}{1 \cdot 3 \cdot 9 + 2 \cdot 6 \cdot 18 + \cdots + n \cdot 3n \cdot 9n}\right)^{1/3}.$$

Qn. 10. In the sketch below $ABCD$ is a square, AX has length 1cm, XY has length 2cm, and YC has length 3cm. Furthermore, $\angle AXY = 90^\circ$ and $\angle XYC = 90^\circ$. Determine the area of the square.



SECTION B : 10 marks each

Qn. 11. E and F are points on the sides CA and AB, respectively, of an equilateral triangle ABC such that EF is parallel to BC. G is the intersection point of medians in triangle AEF and M a point on the segment BE. Prove that $\angle MGC = 60^\circ$ if and only if M is the midpoint of BE.

Qn. 12. Determine all solutions of the following system of equations

$$\begin{aligned} y &= 4x^3 + 12x^2 + 12x + 3 \\ x &= 4y^3 + 12y^2 + 12y + 3 \end{aligned}$$

Qn. 13. In a test, Nabatanzi solved four-fifth of the problems and Amoding solved 35 problems. Half of the problems were solved by both of them. The number of problems solved by neither was a positive one-digit number. What was this number?

Qn. 14. A dog standing at the centre of a circular arena sees a rabbit at the wall. The rabbit runs around the wall and the dog pursues it along a unique path which is determined by running at the same speed and staying on the radial line joining the centre of the arena to the rabbit. Show that the dog overtakes the rabbit just as it reaches a point one-quarter of the way around the arena.

Qn. 15. The non-zero numbers a, b, c, x, y and z are such that $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$. Determine the value of

$$\frac{xyz(a+b)(b+c)(c+a)}{abc(x+y)(y+z)(z+x)}.$$

NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 1.

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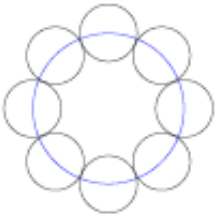
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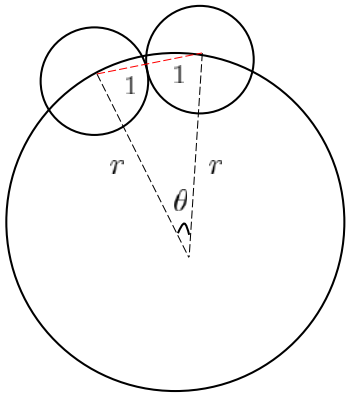
(f) *ALL participants who qualified for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 28*th July,* 2018 *at 8:00am at Makerere University, Kampala.*

SECTION A : 5 **marks each**



Qn. 1. Eight circles of radius 1cm have centers on a larger common circle and adjacent circles are tangent. Find the area of the common circle.

Solution: For r, the radius of the big circle, we have



$$\theta = \frac{360}{8} = \frac{\pi}{4}$$

applying the cosine rule

$$2^2 = r^2 + r^2 - 2r^2\cos\left(\frac{\pi}{4}\right) \Rightarrow r^2 = \frac{4}{2 - 2\cos\left(\frac{\pi}{4}\right)} = \frac{4}{2-\sqrt{2}}$$

Therefore, the area of the big circle is given by

$$A = \pi r^2 = \frac{4\pi}{2-\sqrt{2}} = \frac{4\pi}{2-\sqrt{2}} \cdot \frac{2+\sqrt{2}}{2+\sqrt{2}} = 4\pi\left(\frac{2+\sqrt{2}}{2}\right) = 2\pi\left(2+\sqrt{2}\right)$$

Qn. 2. Determine all the real solutions of the following equation:

$$2^{(2^x)} - 3 \cdot 2^{\left(2^{x-1}+1\right)} + 8 = 0.$$

Solution: Let $y = 2^{x-1}$. Then $2^x = 2y$. Then the left-hand side of the equation becomes

$$2^{2y} - 3 \cdot 2^{y+1} + 8 = 0.$$

Equivalently,

$$2^{2y} - 6 \cdot 2^y + 8 = 0.$$

This factors as

$$(2^y - 4)(2^y - 2) = 0.$$

Therefore, $2^y = 2$ or 4. This yields the solutions $y = 1, 2$. Therefore, $2^{x-1} = 1, 2$, which yields solutions $x - 1 = 0, 1$. Hence, $x = 1, 2$.

We now verify these are indeed solutions. If $x = 1$, then

$$2^{2^1} - 3 \cdot 2^{2^0+1} + 8 = 2^2 - 3 \cdot 2^2 + 8 = 4 - 12 + = 0.$$

Hence, $x = 1$ is a solution. If $x = 2$, then

$$2^{2^2} - 3 \cdot 2^{2^1+1} + 8 = 2^4 - 3 \cdot 2^3 + 8 = 16 - 24 + 8 = 0.$$

Hence, the solutions are indeed $x = 1, 2$.

Qn. 3. Compute the integer k, $k > 2$, for which

$$\log_{10}[(k-2)!] + \log_{10}[(k-1)!] + 2 = 2\log_{10}(k!).$$

Solution:

$$\begin{aligned}
\log_{10}[(k-2)!] + \log_{10}[(k-1)!] + 2 &= 2\log_{10}(k!) \\
\log_{10}\left[\frac{k!}{(k-1)(k)}\right] + \log_{10}\left[\frac{k!}{k}\right] + 2 &= 2\log_{10}(k!) \\
\log_{10}[k!] - \log_{10}[(k-1)(k)] + \log_{10}[k!] - \log_{10}[k] + 2 &= 2\log_{10}(k!) \\
\log_{10}[(k-1)(k)] + \log_{10}[k] &= 2 \\
\log_{10}\left[(k-1)(k^2)\right] &= 2 \\
(k-1)(k^2) &= 100 \\
k^3 - k^2 &= 100
\end{aligned}$$

Let

$$\begin{aligned}
f(k) &= k^3 - k^2 - 100 \\
f(0) &= -100 \\
f(1) &= -100 \\
f(2) &= -96 \\
f(3) &= -82 \\
f(4) &= -52 \\
f(5) &= 0
\end{aligned}$$

Therefore, one of the factors is $k = 5$. We have that

$$k^3 - k^2 - 100 = (k-5)\left(k^2 + 4k + 20\right).$$

Since for $k^2 + 4k + 20 = 0$, $b^2 < 4ac$, it violates the condition for $k > 2$. Therefore $k = 5$.

Qn. 4. Let N be a 3-digit number with three distinct non-zero digits. We say that N is *mediocre* if it has the property that when all six 3-digit permutations of N are written down, the average is N. For example, $N = 481$ is mediocre, since it is the average of $418, 481, 148, 184, 814, 841$. Determine the largest mediocre number.

Solution: Suppose abc is a mediocre number. The 6 permutations are $\{abc, acb, bac, bca, cab, cba\}$. The sum of these numbers is $222(a + b + c)$ and the average is $37(a + b + c)$). Since abc is mediocre we have $100a + 10b + c = 37(a + b + c)$.

We can rearrange this equation to get $63a = 27b + 36c$. Notice that $63 = 27 + 36$ so we must have a strictly between b and c. Thus, $a \neq 9$.

Notice if that $b = a + 1$ then we have $36a = 27 + 36c$ which has no integer solutions, and if $c = a + 1$ then we have $27a = 27b + 36$ which has no integer solutions. Since a is strictly between b and c if $a = 8$ then b or c would have to be 9, but this cannot happen. Thus, $a \neq 8$.

If $a = 7$ then either $b = 9$ $c = 9$. This would reduce to $441 = 243 + 36c$ or $441 = 27b + 324$

Neither of these has integer solutions, so $a \neq 7$.

If $a = 6$ then we have $378 = 27b + 36c$ which reduces to $42 = 3b + 4c$. This gives $b = 14 - \frac{4}{3}c$. To get integer digit values we must have $c = 3, 6$, or 9 which gives corresponding b values of $10, 6$, and 2 respectively. We can omit the first two, since 10 is not a digit and 666 is not mediocre. Thus, the largest mediocre number is 629.

Qn. 5. Prove that for all real numbers x, y,

$$\left(x^2+1\right)\left(y^2+1\right)+4\,(x-1)\,(y-1)\geq 0.$$

Determine when equality holds.

Solution: Expanding and refactoring the left hand side of the given inequality gives

$$(xy+1)^2+(x+y-2)^2.$$

This is clearly non-negative, so the inequality holds. This holds with equality when each of the terms is 0. So $xy=-1$ and $x+y=2$. Solving this system yields

$$(x,y)=\left(1-\sqrt{2},\ 1+\sqrt{2}\ \right),\ \ \left(1+\sqrt{2},\ 1-\sqrt{2}\ \right).$$

Qn. 6. If

$$x = \left(1 + \frac{1}{n}\right)^n \quad \text{and} \quad y = \left(1 + \frac{1}{n}\right)^{n+1},$$

show that $y^x = x^y$.

Solution:

$$\begin{aligned}
y^x &= \left(\left(1 + \frac{1}{n}\right)^{n+1}\right)^{\left(1+\frac{1}{n}\right)^n} = \left(1 + \frac{1}{n}\right)^{(n+1)\left(1+\frac{1}{n}\right)^n}; \\
x^y &= \left(\left(1 + \frac{1}{n}\right)^{n}\right)^{\left(1+\frac{1}{n}\right)^{n+1}} \\
&= \left(1 + \frac{1}{n}\right)^{n\left(1+\frac{1}{n}\right)^{n+1}} \\
&= \left(1 + \frac{1}{n}\right)^{n\left(1+\frac{1}{n}\right)^{n}\left(1+\frac{1}{n}\right)} \\
&= \left(1 + \frac{1}{n}\right)^{(n+1)\left(1+\frac{1}{n}\right)^{n}}
\end{aligned}$$

Qn. 7. Find all number triples (x, y, z) such that when any of these numbers is added to the product of the other two, the result is 2.

Solution: The system to be solved is:

$$x + yz = 2; \quad y + zx = 2; \quad z + xy = 2.$$

Subtracting the second from the first and the third from the second yields, respectively,

$$(x-1)(1-z) = 0, \quad (y-z)(1-x) = 0.$$

Each of the four cases

$$\begin{aligned} x - y &= 0 = y - z, \quad x - y = 0 = 1 - x, \\ 1 - z &= 0 = y - z \quad 1 - z = 0 = 1 - x, \end{aligned}$$

implies

$$x = y = z = 1$$

or

$$x = y = z = -2$$

Qn. 8. Suppose that n people each know exactly one piece of information, and all n pieces are different. Every time person A phones person B, A tells B everything that A knows, while B tells A nothing. What is the minimum number of phone calls between pairs of people needed for everyone to know everything?

Solution: Let us denote the n people by $p_1, p_2, \ldots, p_n$, and a call from p_i to p_j by $p_i \to p_j$.

Let $f(n)$ denote the minimum number of (one-way) calls which can leave everybody fully informed. The particular sequence

$$p_1 \to p_n, p_2 \to p_n, \ldots, p_{n\cdot 1} \to p_n, p_n \to p_{n\cdot 1}, p_n \to p_{n\cdot 2}, \ldots, p_n \to p_1$$

contains $2n-2$ calls and leaves everybody informed, showing that $f(n) \leq 2n-2$. Now suppose we have a given sequence of calls which leaves everybody fully informed.

Consider the "crucial" call at the end of which the receiver (call him p) is fully informed, but no one else is. Clearly each of the $n-1$ people (other than p) must have placed at least one call no later than this crucial call (how else could p be fully informed?) and each of these $n-1$ people (being not fully informed at the end of the crucial call) must receive at least one call after the crucial one. Hence the given sequence contains at least $2(n-1)$ calls. Hence

$$f(n) = 2n-2.$$

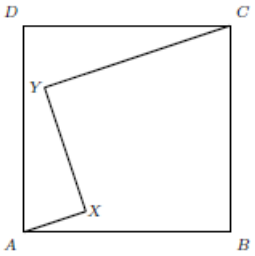
Qn. 9. Simplify

$$\left(\frac{1\cdot 2\cdot 4+2\cdot 4\cdot 8+\ldots+n\cdot 2n\cdot 4n}{1\cdot 3\cdot 9+2\cdot 6\cdot 18+\cdots+n\cdot 3n\cdot 9n}\right)^{1/3}$$

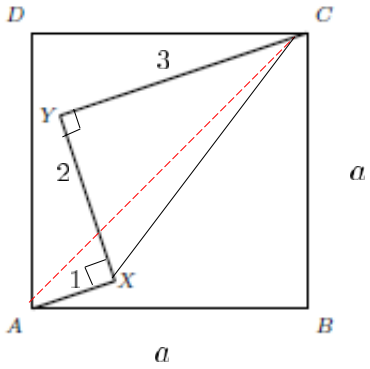
Solution: Since $k\cdot 2k\cdot 4k=8k^3$ and $k\cdot 3k\cdot 9k=27k^3$ for $1\leq k\leq n$,

$$\begin{aligned}\left(\frac{1\cdot 2\cdot 4+2\cdot 4\cdot 8+\ldots+n\cdot 2n\cdot 4n}{1\cdot 3\cdot 9+2\cdot 6\cdot 18+\ldots+n\cdot 3n\cdot 9n}\right)^{1/3} &= \left(\frac{8\left[1^3+2^3+\ldots+n^3\right]}{27\left[1^3+2^3+\ldots+n^3\right]}\right)^{1/3}\\ &= \left(\frac{8}{27}\right)^{1/3}\\ &= \frac{2}{3}.\end{aligned}$$

Qn. 10. In the sketch below $ABCD$ is a square, AX has length 1cm, XY has length 2cm, and YC has length 3cm. Furthermore, $\angle AXY = 90°$ and $\angle XYC = 90°$. Determine the area of the square.



Solution: 10 From the figure



$$\begin{aligned} AX &= 1 \\ XY &= 2 \\ YC &= 3 \\ XC &= \sqrt{2^2+3^2} = \sqrt{13} \\ \angle YXC &= \sin^{-1}\frac{3}{\sqrt{13}} \\ \angle AXC &= 90° + \sin^{-1}\frac{3}{\sqrt{13}} \end{aligned}$$

By Cosine rule,

$$\begin{aligned} AC &= \sqrt{1^2 + \left(\sqrt{13}\right)^2 - 2(1)\left(\sqrt{13}\right)\cdot\cos\left(90° + \sin^{-1}\frac{3}{\sqrt{13}}\right)} \\ &= \sqrt{1^2 + \left(\sqrt{13}\right)^2 + 2(1)\left(\sqrt{13}\right)\cdot\sin\left(\sin^{-1}\frac{3}{\sqrt{13}}\right)} \\ &= \sqrt{20} \\ &= 2\sqrt{5} \end{aligned}$$

Using $\triangle ABC$ and the Pyhtogras theorem,

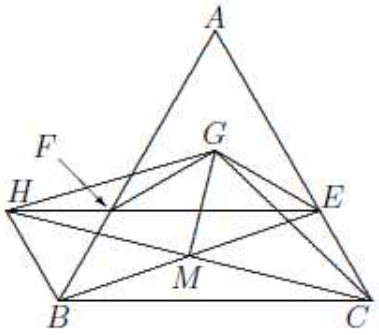
$$\begin{aligned} a^2 + a^2 &= AC^2 \\ 2a^2 &= AC^2 \quad \Rightarrow \\ a &= \frac{AC}{\sqrt{2}} = \frac{2\sqrt{5}}{\sqrt{2}} = \sqrt{2}\sqrt{5} = \sqrt{10} \end{aligned}$$

The area of the square is then given by $A = a \times a = a^2 = \left(\sqrt{10}\right)^2 = 10$ square units.

SECTION B : 10 marks each

Qn. 11. E and F are points on the sides CA and AB, respectively, of an equilateral triangle ABC such that EF is parallel to BC. G is the intersection point of medians in triangle AEF and M a point on the segment BE. Prove that $\angle MGC = 60^\circ$ if and only if M is the midpoint of BE.

Solution: Let H be on the extension of EF such that $BCEH$ is a parallelogram. Then $\angle BHE = \angle BCE = 60^\circ$ and $\angle BFH = \angle AFE = 60^\circ$, so FBH is also an equilateral triangle, and in particular $HF = FB = EC$. Also $HFG = 180^\circ - \angle GFE = 180^\circ - 30^\circ = 150^\circ$, and similarly $\angle CEG = 150^\circ$, thus $\angle HFG = \angle CEG$.



Since $\angle GFE = 30^\circ = \angle GEF$, we know $FG = GE$. Hence triangles HFG and CEG are congruent (by SAS) so that $GH = GC$ and $\angle HGF = \angle CGE$. It follows that $\angle HGC = \angle HGF + \angle FGC = \angle CGE + \angle FGC = \angle EGF = 180^\circ - \angle GFE - \angle GEF = 180^\circ - 30^\circ - 30^\circ = 120^\circ$.

Now M is the midpoint of the diagonal BE of the parallelogram if and only if it is the midpoint of the diagonal HC. Since triangle HGC is isosceles, this is equivalent to GM being the bisector of $\angle HGC$. In other words, $\angle MGC = 60^\circ$.

Qn. 12. Determine all solutions of the following system of equations

$$\begin{aligned} y &= 4x^3 + 12x^2 + 12x + 3 \\ x &= 4y^3 + 12y^2 + 12y + 3 \end{aligned}$$

Solution: We can rewrite the two equations as:

$$\begin{aligned} y + 1 &= 4(x+1)^3 \\ x + 1 &= 4(y+1)^3 \end{aligned}$$

Substituting the second equation into the first, we get

$$(y+1) = 256(y+1)^9$$

By inspection we see that $y = -1$ satisfies the equation. Assume $y \neq -1$, then the equation becomes:

$$\frac{1}{2^8} = (y+1)^8$$

The only real solutions to this are $y + 1 = \pm\frac{1}{2}$.

This gives $y = -\frac{1}{2}, -1, -\frac{3}{2}$ as possible solutions.

Substituting into the above equations, we get the following three solutions:

$$\left(-\frac{1}{2}, -\frac{1}{2}\right), \ (-1, -1), \ \left(-\frac{3}{2}, -\frac{3}{2}\right).$$

Qn. 13. In a test, Nabatanzi solved four-fifth of the problems and Amoding solved 35 problems. Half of the problems were solved by both of them. The number of problems solved by neither was a positive one-digit number. What was this number?

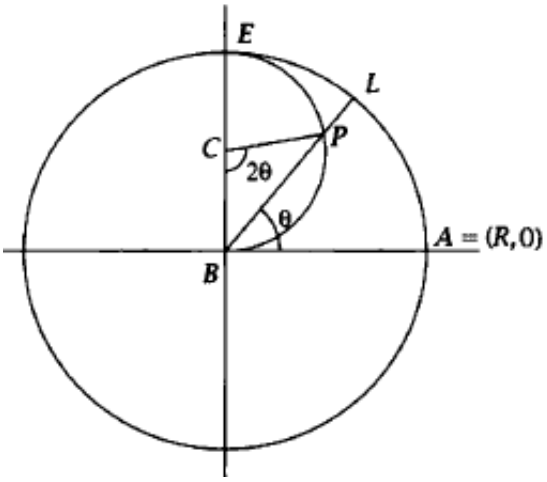
Solution: The fraction of problems solved only by Nabatanzi was $\frac{4}{5} - \frac{1}{2} = \frac{3}{10}$ so that the total number of problems is a multiple of 10. The fraction of problems solved by Amoding was at most $1 - \frac{3}{10} = \frac{7}{10}$. Thus the total number of problems was at least 50.

If it were 50, then 10 problems were solved by Amoding alone, and as Nabatanzi solved $\frac{4}{5} \times 50 = 40$ problems, the number of problems solved by neither is 0. The total number of problems could only be as large as 70 since 35 problems would be solved by both. In this case, the number of problems solved by neither was $\frac{1}{5} \times 70 = 14$.

It follows that the total number of problems must be 60, of which 30 were solved by both, 5 by Amoding alone, $\frac{3}{10} \times 60 = 18$ problems by Nabatanzi alone, and $60 - 30 - 5 - 18 = 7$ by neither of them.

Qn. 14. A dog standing at the centre of a circular arena sees a rabbit at the wall. The rabbit runs around the wall and the dog pursues it along a unique path which is determined by running at the same speed and staying on the radial line joining the centre of the arena to the rabbit. Show that the dog overtakes the rabbit just as it reaches a point one-quarter of the way around the arena.

Solution a: Let the centre of the circular arena be $B(0,0)$ and the radius R. Assume that the rabbit runs counterclockwise starting from $A(R,0)$. We shall show that if the dog runs along a circular arc with centre $C\left(0,\frac{R}{2}\right)$ and radius $\frac{R}{2}$, such that at time t its speed is the same as that of the rabbit, then the dog and the rabbit are on the same radial line. This will show that the circle is a possible path for the dog. The uniqueness statement given in the problem implies that this is the only path.



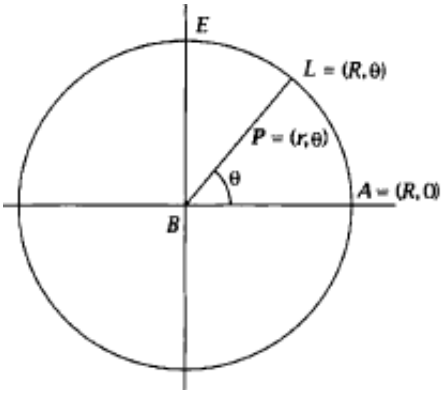
Since the dog and the rabbit are on the same radial line, they meet at the required point $E(0,R)$. Let L be the position of the rabbit at any time t and P be the point where BL meets the circle with centre C and radius $\frac{R}{2}$. It is sufficient to show that the arcs AL and BP are equal in length, since this implies that P is the position of the dog at time t.

Let $ABL=\theta$. Then angle $BCP=2\theta$ since angle $BPE=\frac{\pi}{2}$. Hence, arc $AL=R\theta$ and arc $BP=\frac{R}{2}(2\theta)=R\theta$. This completes the proof.

Solution b: Let the rabbit begin at $A(R,0)$ and the dog at $B(0,0)$. At any moment, suppose the dog is at P and the rabbit is at L. We are given that at any time t,

(a) $B, P,$ and L are collinear, and

(b) the speed of P is the same as of L.



Let the position of the dog be given in polar coordinates as $P(r,\theta)$.
Then condition (a) implies:

$$\text{speed of the dog} = \sqrt{\left(\frac{dr}{dt}\right)^2 + r^2\left(\frac{d\theta}{dt}\right)^2}$$

$$\text{speed of the rabbit} = \left(R\frac{d\theta}{dt}\right)$$

Condition (b) implies

$$\begin{aligned}\left(\frac{dr}{dt}\right)^2 + r^2\left(\frac{d\theta}{dt}\right)^2 &= R^2\left(\frac{d\theta}{dt}\right)^2, \quad \text{or} \\ \left(\frac{dr}{dt}\right)^2 &= (R^2 - r^2)\left(\frac{d\theta}{dt}\right)^2\end{aligned}$$

Hence,

$$\frac{dr}{\sqrt{R^2 - r^2}} = d\theta.$$

Integrating each side gives $\arcsin\dfrac{r}{R} = \theta + C$. Hence, $r = R\sin(\theta + c)$. Initially, $r = 0$ and $\theta = 0$ so $c = 0$. The path traced by the dog, is therefore, $r = R\sin\theta$ which is a circle of radius $\dfrac{R}{2}$ and centre $\left(0, \dfrac{R}{2}\right)$. This completes the proof.

Qn. 15. The non-zero numbers a, b, c, x, y and z are such that $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$. Determine the value of

$$\frac{xyz(a+b)(b+c)(c+a)}{abc(x+y)(y+z)(z+x)}$$

Solution: Let $\frac{x}{a} = \frac{y}{b} = \frac{z}{c} = r$. Then

$$\begin{aligned} \frac{xyz}{abc} &= r^3 \\ x &= ra \\ y &= rb \quad \text{and} \\ z &= rc \end{aligned}$$

It follows that $x + y = r(a + b),\ y + z = r(b + c)$ and $z + x = r(c + a)$, so that

$$\frac{x+y}{a+b} = \frac{y+z}{b+c} = \frac{z+x}{c+a} = r.$$

The given expression is equal to

$$\frac{xyz(a+b)(b+c)(c+a)}{abc(x+y)(y+z)(z+x)} = r^3 \left(\frac{1}{r}\right)^3 = 1$$